Battery Arbitrage Optimization
Mathematical Formulation

1 Mathematical formulation

This section describes the mixed-integer linear programming (MILP) model used to optimize battery trading across multiple electricity markets with different trading interval durations.

The model maximizes trading cash flow subject to the battery's power and energy limits. Market commitments are defined at their respective market interval resolutions, while the battery's energy balance and operating mode are represented on a common physical timeline.

The formulation permits negative electricity prices. Prices are treated as input parameters and are not constrained to be nonnegative.

1.1 Sets and indices

- $\mathcal{M}$: set of markets, indexed by m.
- $\mathcal{T} = \{0, \ldots, N-1\}$: set of physical timesteps, each of duration Δ hours.
- $s_m \in \mathbb{N}$: number of physical timesteps in one trading interval of market m.
- $\mathcal{K}_m = \{0, \ldots, N/s_m - 1\}$: set of trading intervals for market m. The model requires $s_m \mid N$, so the optimization horizon covers complete intervals for every market.
- $\kappa_m(t) = \lfloor t/s_m \rfloor$: index of the market-$m$ interval containing physical timestep t.

The physical timeline begins at a boundary shared by all markets. Each market interval must be an integer multiple of the physical timestep.

For example, with a half-hourly physical timeline, $s_1 = 1$ represents a half-hourly market, whereas $s_2 = 2$ represents an hourly market.

1.2 Parameters and decision variables

The model uses the following parameters:

- $p_{m,t}$: price of market m at physical timestep t, measured in £/MWh. This parameter is obtained by aligning each market's interval prices to the common physical timeline.
- Δ: physical timestep duration, in hours.
- P^c: maximum charging power, in MW.
- P^d: maximum discharging power, in MW.
- $E^{\max}$: maximum stored energy, in MWh.
- E_0: initial stored energy, in MWh.

The decision variables are:

$$
\begin{aligned}
c_{m,k} &\in [0, P^c], && m \in \mathcal{M},\ k \in \mathcal{K}_m, && (1)\\
d_{m,k} &\in [0, P^d], && m \in \mathcal{M},\ k \in \mathcal{K}_m, && (2)\\
e_t &\in [0, E^{\max}], && t = 0, \ldots, N, && (3)\\
z_t &\in \{0, 1\}, && t \in \mathcal{T}. && (4)
\end{aligned}
$$

Here:

- $c_{m,k}$ is the charging power committed in market m during interval k, in MW.
- $d_{m,k}$ is the discharging power committed in market m during interval k, in MW.
- e_t is the stored energy at the start of physical timestep t, in MWh.
- z_t represents the battery's operating mode: $z_t = 1$ permits charging, while $z_t = 0$ permits discharging.

The initial energy is fixed at

$$e_0 = E_0. \tag{5}$$

Charging and discharging decisions are indexed by market interval rather than physical timestep. Consequently, a commitment is constant throughout the market interval it covers. The energy state and operating mode, however, are defined at the finer physical-timestep resolution.

1.3 Optimization model

Define the aggregate charging and discharging powers at each physical timestep as

$$
\begin{aligned}
C_t &= \sum_{m \in \mathcal{M}} c_{m,\kappa_m(t)}, && t \in \mathcal{T}, && (6)\\
D_t &= \sum_{m \in \mathcal{M}} d_{m,\kappa_m(t)}, && t \in \mathcal{T}. && (7)
\end{aligned}
$$

The optimization model is

$$
\begin{aligned}
\max \quad & \sum_{m \in \mathcal{M}} \sum_{t \in \mathcal{T}} \left(d_{m,\kappa_m(t)} - c_{m,\kappa_m(t)} \right) p_{m,t} \Delta && && (8)\\
\text{subject to} \quad & e_{t+1} = e_t + \Delta(C_t - D_t), && t \in \mathcal{T}, && (9)\\
& C_t \le P^c z_t, && t \in \mathcal{T}, && (10)\\
& D_t \le P^d (1 - z_t), && t \in \mathcal{T}, && (11)\\
& e_0 = E_0, && && (12)\\
& 0 \le e_t \le E^{\max}, && t = 0, \ldots, N, && (13)\\
& 0 \le c_{m,k} \le P^c, && m \in \mathcal{M},\ k \in \mathcal{K}_m, && (14)\\
& 0 \le d_{m,k} \le P^d, && m \in \mathcal{M},\ k \in \mathcal{K}_m, && (15)\\
& z_t \in \{0, 1\}, && t \in \mathcal{T}. && (16)
\end{aligned}
$$

The objective in (8) maximizes total trading cash flow, calculated as sales revenue minus electricity purchase costs. Charging and discharging powers are multiplied by the corresponding market price and timestep duration to obtain cash flow in pounds.

The energy balance in (9) tracks the battery's stored energy on the shared physical timeline. It assumes 100% charging and discharging efficiency, with no additional losses.

The operating-mode constraints in (10)–(11) prevent simultaneous charging and discharging across all markets at each physical timestep.

The power ratings P^c and P^d act as tight big-M constants in these operating-mode constraints. Since aggregate charging and discharging are bounded by the corresponding power ratings, the constraints enforce mutual exclusivity without unnecessarily large big-M constants.

The terminal energy e_N is bounded by the storage capacity but is otherwise unconstrained. In particular, the model does not require the battery to return to its initial energy level.

1.4 Properties and assumptions

Market commitments and cash-flow accounting. Market commitments remain constant throughout their respective intervals. The objective sums cash flow over physical timesteps.

The formulation assumes that the alignment stage repeats each market's interval price across the physical timesteps it covers. If $p_{m,t} = p_{m,k}$ for all physical timesteps t covered by market interval k, that interval's contribution to the objective is

$$(d_{m,k} - c_{m,k})p_{m,k}s_m\Delta. \tag{17}$$

Thus, the interval's cash flow equals its net committed power multiplied by its price and interval duration.

The optimizer relies on this price-alignment property but does not independently validate price constancy within each market interval.

Negative electricity prices. The model does not impose a non-negativity constraint on $p_{m,t}$. Negative prices are therefore valid inputs to the optimization model.

A negative price changes the economics of charging and discharging. For example, when $p_{m,t} < 0$, charging at a positive power contributes positively to the objective because the charging term is $-c_{m,\kappa_m(t)}p_{m,t}\Delta$.

For a charging commitment of 5 MW over half an hour at a price of $-£20$/MWh, the charging cash-flow contribution is

$$-5 \times (-20) \times 0.5 = £50. \tag{18}$$

This is a simplified example under the model's settlement assumptions. It does not account for transaction costs, market-specific settlement rules, or other operational charges.

The optimizer considers negative prices together with the energy balance, power limits, and storage capacity. Consequently, it will not necessarily charge whenever prices are negative: the decision also depends on the remaining horizon, available capacity, future prices, and the feasibility of subsequent actions.

Shared power and energy constraints. All market commitments contribute to the aggregate charging and discharging powers C_t and D_t. Consequently, the battery's power limits apply to its total activity across markets rather than independently to each market.

The shared energy balance and storage bounds ensure that the same stored energy cannot be sold twice. These constraints are essential when market commitments have different interval durations and overlap on the physical timeline.

Mutual exclusivity across markets. If any market has a positive charging commitment during a physical timestep, the aggregate charging constraint requires $z_t = 1$. The aggregate discharging constraint then forces $D_t = 0$.

Conversely, positive discharging requires $z_t = 0$ and prevents charging. Because a market commitment spans all its covered physical timesteps, charging and discharging cannot both be positive within the same market interval.

Why binary variables are needed. Without the operating-mode constraints, the model could simultaneously purchase electricity in one market and sell it in another during the same physical timestep. If the purchase price were lower than the selling price, this could generate positive cash flow without a corresponding net change in stored energy.

The binary operating-mode variables prevent this behavior and make the formulation a mixed-integer linear program (MILP), rather than a linear program (LP).

Feasibility. For any valid battery configuration and valid timeline, the model has a feasible solution: set all charging and discharging commitments to zero and keep the stored energy equal to E_0 throughout the horizon.

The configuration validation ensures that $0 \leq E_0 \leq E^{\max}$, while timeline validation checks the time-grid and market-interval requirements. The solver wrapper's infeasibility handling is therefore defensive, assuming these validations correctly capture the model's configuration requirements.

Initial and terminal energy effects. The default initial energy is $E_0 = 0$ MWh. The model assigns no terminal value to stored energy and does not impose a terminal energy target. Therefore, the resulting objective does not necessarily correspond to a complete cycle that returns the battery to its initial state.

When prices are nonnegative, leaving purchased energy unused at the end generally offers no additional benefit under this objective, although zero-price charging can produce alternative optimal terminal states. When prices are negative, charging can itself generate positive cash flow, so the optimal solution may end with energy remaining in the battery.

The model can also discharge energy present at the beginning of the horizon without requiring that energy to be replenished before the end.

These effects follow from the selected horizon and terminal-state assumptions rather than from errors in the energy balance. The reported objective represents optimized trading cash flow over the selected period under these assumptions. It does not include a terminal valuation for residual stored energy.

Scope and simplifications. The formulation assumes perfect foresight of market prices over the optimization horizon. It does not include battery degradation, transaction costs, charging

or discharging efficiency losses, ramping constraints, or market-specific restrictions beyond those represented by the decision variables and shared physical constraints.

These assumptions simplify the optimization problem and should be considered when interpreting the resulting trading cash flow.